Nash equilibria of quasisupermodular games

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Abstract

We prove three results on the existence and structure of Nash equilibria for quasisupermodular games. A theorem is purely order-theoretic, and the other two involve topological hypotheses. Our topological results generalize Zhou's theorem (for supermodular games) and Calciano's theorem.

Keywords— Quasisupermodular games, Nash equilibrium, Tarski fixed point theorem

1 Introduction

Quasisupermodular games, introduced by Shannon and Milgrom in [17] and [13], are generalizations of supermodular games. Supermodular games exhibit "strategic complementarities", wherein the choices of one player positively influence the preferences of others, leading them to favor similar higher actions in response.

The defining properties of supermodular games, namely supermodularity and increasing difference property for *real-valued* payoff functions, rely essentially on the additive group structure of the *codomain* $\mathbb{R}$. Milgrom and Shannon [13, p.160, p.162] generalize the two conditions to quasisupermodularity (Definition 4.1) and single crossing property (Definition 6.1). Their conditions require no longer any additive structure on the codomain, so extend to chain-valued functions ([20, p.59]). Quasisupermodular games retain the main feature of supermodular games, i.e., complementarities among actions. The scope of quasisupermodular games surpasses that of supermodular games, broadening the spectrum of economic scenarios to which it can be applied.

Assuming order upper semicontinuity on the payoff functions, Milgrom and Roberts [12, p.1266] show that a supermodular game has a least and a largest Nash equilibrium. Under (topological) continuity conditions on the payoff functions, Milgrom and Shannon [13, Theorem 12] give a similar result for quasisupermodular games. Then the same conclusion holds even if the continuity restriction is relaxed and the payoff functions are allowed to be more general poset-valued functions ([3, Proposition 8.49]). Here and below, a set equipped with a partial order is called a poset.

The study on the structure of the set of Nash equilibria of supermodular games is initialized by Vives [22, Theorem 4.2]. Veinott [21, Ch. 10] establishes the existence

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and studies the structure of Nash equilibria for parameterized games with certain complementarities, which are variants of quasisupermodular games. Zhou [24, Theorem 2] shows that for a supermodular game, the Nash equilibrium set with the inherited order structure forms a complete lattice. Zhou's result is extended to quasisupermodular games by Calciano [2, Theorem 3].

The main objective of this paper is twofold: the first concerns (pure) Nash equilibria of quasisupermodular (normal form) games. We provide a *purely order-theoretic* result (Theorem 6.8) concerning the structure of the set of Nash equilibria. As opposed to Nash's theorem on the existence of Nash equilibria, it does not require any topological hypothesis. The proof of Theorem 6.8 needs an existence result of maximum (Theorem 5.3). Second, in the spirit of [15, Theorem 1], we relax the conventional topological condition (upper semicontinuity) of the payoff functions to transfer upper continuity (Definition 3.1) in Theorems 6.11 and 6.13.

The text is organized as follows. Section 2 summarizes two completeness properties for lattices and shows their equivalence. Section 3 recalls several conditions analogous to continuity, both in the topological or the order-theoretic sense. Section 4 reviews quasisupermodularity. Section 5 gives existence results about maximum for functions. Finally in Section 6, results from the previous sections are used to establish the existence of Nash equilibria for quasisupermodular games. The order structure of the set of all Nash equilibria is also discussed.

2 A result of Veinott

In Section 2, we give a self-contained proof of a theorem due to Veinott [21, Ch. 2] (recalled as Theorem 2.3). It is used in the proof of Theorem 6.8, and we cannot find a published proof in the literature.

A poset L is called a lattice if for any $x, y \in L$, both $\sup_L\{x, y\}$ and $\inf_L\{x, y\}$ exist in L, classically denoted by $x \vee y$ and $x \wedge y$ respectively. A poset L is called a complete lattice if every nonempty subset A of L, both $\sup_L(A)$ and $\inf_L(A)$ exist. (Such an L is necessarily a lattice.) A poset is called a chain (resp. an anti-chain) if any two distinct elements are comparable (resp. incomparable). In an anti-chain with more than one element, every element is minimal but not the least element. For a lattice, Lemma 2.1 shows the two notions agree.

Lemma 2.1. *A minimal (resp. maximal) element of a lattice is the least (resp. largest) element.*

Proof. Let $s \in L$ be a minimal element of a lattice L. For every $s' \in L$, the element $s \wedge s'$ exists in L. Since s is minimal and $s \wedge s' \leq s$, one has $s = s \wedge s' \leq s'$, which shows that s is the least element. The statement in the parentheses is similar. □

Definition 2.2. Let P be a poset. Let S be a subset of P. If for every nonempty chain (resp. subset) $C \subset S$, both $\sup_P(C)$ and $\inf_P(C)$ exist and lie in S, then S is called chain-subcomplete (resp. subcomplete) in P. If P is chain-subcomplete in itself, then P is called chain-complete.

Topkis [20, p.29] gives a complete sublattice that is not subcomplete.

In [10, Theorem B], Theorem 2.3 below is stated without proof and attributed to Veinott's unpublished work [21]. Veinott's proof relies on transfinite induction. We provide a new proof for the reader's convenience.

Theorem 2.3 (Veinott)**.** *Let S be a sublattice of a poset P. Then S is subcomplete if and only if it is chain-subcomplete in P.*

Proof. The "only if" part follows from the definition. We prove the "if" part.

Suppose to the contrary that S is not subcomplete. Then there is a nonempty subset $A \subset S$ such that $\sup_P(A)$ or $\inf_P(A)$ is not in S. By symmetry, we may assume that $\sup_P(A)$ is not in S. As the *class* of cardinal numbers is well-ordered (see, e.g., [8]), we may assume that A is of minimal cardinality among all such subsets. Since S is a lattice, the set A is infinite. Then by [16, Thm. 2.17], there is a chain I and for every $\alpha \in I$ a subset $A_\alpha \subset A$ with $|A_\alpha| < |A|$, such that $A = \cup_{\alpha \in I} A_\alpha$ and for any $\alpha < \beta$ in I, one has $A_\alpha \subset A_\beta$.

By the minimality of $|A|$, for every $\zeta \in I$, the element $b_\zeta := \sup_P(A_\zeta)$ exists and is in S. For any two such indices $\zeta \leq \eta < |A|$, one has $A_\zeta \subset A_\eta$, so $b_\zeta \leq b_\eta$. Hence $\{b_\alpha\}_{\alpha \in I}$ is a nonempty increasing chain in S. Since $S \subset P$ is chain-subcomplete, the element $b = \sup_P\{b_\alpha\}_{\alpha \in I}$ exists and is in S.

We claim that $b = \sup_P(A)$. For every $a \in A$, there is $\alpha_0 \in I$ with $a \in A_{\alpha_0}$, so $a \leq b_{\alpha_0} \leq b$. If $b' \in P$ is such that $b' \geq a$ for all $a \in A$, then $b' \geq b_\alpha$ for every $\alpha \in I$. Therefore, $b' \geq b$. The claim is proved.

However, the claim contradicts the choice of A. This contradiction finishes the proof. □

Remark 2.4. Theorem 2.3 implies [4, Thm. 2.41 (iii)]: a lattice having no infinite chains is complete.

3 Continuity

In Section 3, we review some variants of the continuity in topology. They are different from the conventional upper semicontinuity, but suffice to ensure the existence of a maximum. They are part of the hypotheses of our main results in Section 6.

Transfer continuity conditions are introduced by Tian and Zhou [19, Definitions 1, 2]. They are used to establish the existence of Nash equilibria in discontinuous games in [14].

Definition 3.1. Let X be a topological space. A function $f : X \to \mathbb{R}$ is called transfer (resp. weakly transfer) upper continuous if for any $x, y \in X$ with $f(y) < f(x)$, there exists a point $x' \in X$ and an open neighborhood U of $y \in X$ such that $f(z) < f(x')$ (resp. $f(z) \leq f(x')$) for every $z \in U$.

An upper semicontinuous function is transfer upper continuous, and a transfer upper continuous is transfer weakly upper continuous. As [19, Examples 1,2] show, the converses fail.

We recall several continuity involving order. Let $\mathscr{C}$ (resp. $(L, \leq)$) be a chain (resp. complete lattice). Let $f : L \to \mathscr{C}$ be a function. It is called *topologically upper semicontinuous*, if for every $a \in \mathscr{C}$, the set $[f \geq a] := \{x \in L : f(x) \geq a\}$ is closed in the interval topology of L. (When $\mathscr{C} = \mathbb{R}$, this reduces to the upper semicontinuity of real-valued functions.)

Definition 3.2 is an order-theoretic counterpart of the topological semicontinuity. We review the notation from [12, p.1261]. For a nonempty chain $C \subset L$, the pair $(C, \leq)$ is naturally a directed set. Then the inclusion $C \to L$ is a net in L, which is denoted by $x \in C, x \uparrow \sup_L(C)$. Define the reversed order $\preceq$ on C, by $x \preceq y$ if and only if $x \geq y$ for

all $x, y \in C$. Then $(C, \preceq)$ is another directed set. The inclusion $C \to L$ gives another net in L, which is denote by $x \in C, x \downarrow \inf_L(C)$. For every net $x_\bullet : (I, \leq) \to L$, define the limit superior along the net by $\limsup_{i \in I} x_i := \inf_{i \in I} \sup_{j \geq i} x_j$. It is well-defined by completeness of L.

Definition 3.2 ([12, p.1260], [15, Definition 1])**.** Assume that the chain $\mathscr{C}$ is complete. The function f is *upward* (resp. *downward*) *upper semicontinuous* if for every nonempty chain $C \subset L$, one has

$$\limsup_{x \in C, x \uparrow \sup_L(C)} f(x) \leq f(\sup_L(C))$$

$$(\text{resp.} \limsup_{x \in C, x \downarrow \inf_L(C)} f(x) \leq f(\inf_L(C))).$$

If f is both upward and downward upper semicontinuous, then it is called *order upper semicontinuous.*

Remark 3.3. For a finite chain C, the corresponding inequalities in Definition 3.2 are automatic. In particular, if every chain in L is finite, then every function on L is order upper semicontinuous. The sum of two upward (resp. downward) upper semicontinuous functions on L is also upward (resp. downward) upper semicontinuous.

We shall give a result (Theorem 6.8) concerning the existence of Nash equilibria of quasisupermodular games, assuming only the continuity in Definition 3.4.

Definition 3.4. [17, p.8] The function $f : L \to \mathscr{C}$ is *upper chain subcomplete*, if for every $a \in \mathscr{C}$ and every nonempty chain C in $[f \geq a]$, both $\sup_L(C)$ and $\inf_L(C)$ are in $[f \geq a]$.

Definition 3.4 is an extension of Definition 3.2 to non-complete chains.

Proposition 3.5. *Assume that the chain $\mathscr{C}$ is complete. Then f is order upper semicontinuous if and only if f is upper chain subcomplete.*

Proof. We prove that f is upward upper semicontinuous if and only if for every $a \in \mathscr{C}$ and every nonempty chain C in $[f \geq a]$, one has $\sup_L(C) \in [f \geq a]$.

Assume that f is upward upper semicontinuous. Then for every $a \in \mathscr{C}$ and every nonempty chain C in $[f \geq a]$, one has

$$f(\sup_L(C)) \geq \limsup_{x \in C, x \uparrow \sup_L(C)} f(x) \geq a,$$

so $\sup_L(C) \in [f \geq a]$.

Conversely, assume that for every $a \in \mathscr{C}$ and every nonempty chain C' in $[f \geq a]$, one has $\sup_L(C') \in [f \geq a]$. For a nonempty chain $C \subset S$, we prove

$$f(\sup_L(C)) \geq \limsup_{x \in C, x \uparrow \sup_L(C)} f(x). \tag{1}$$

Otherwise, one has $f(\sup_L(C)) < \limsup_{x \in C, x \uparrow \sup_L(C)} f(x)$. Because $\mathscr{C}$ is a chain, there are exactly two cases.

1. There is $b \in \mathscr{C}$ with $f(\sup_L(C)) < b < \limsup_{x \in C, x \uparrow \sup_L(C)} f(x)$.
2. For every $\beta \in \mathscr{C}$ with $\beta > f(\sup_L(C))$, one has $\beta \geq \limsup_{x \in C, x \uparrow \sup_L(C)} f(x)$.

In Case 2, set $b = \limsup_{x\in C, x\uparrow \sup_L(C)} f(x)$. Then for every $x \in C$, one has $\sup_{y\in C, y\geq x} f(y) \geq b > f(\sup_L(C))$. There is $y_x (\geq x)$ in C with $f(y_x) > f(\sup_L(C))$. Then $f(y_x) \geq b$.

In both cases, there is a directed set I and an order-preserving function $h : I \to C$ (i.e., a Willard-subnet) such that $h(I)$ is a cofinal subset of C and for every $u \in I$, one has $f(h(u)) \geq b$. Then $h(I) \subset [f \geq b]$.

Since $h(I) \subset C$, one has $\sup_L(C) \geq \sup_L(h(I))$. For every $x \in C$, as $h(I)$ is cofinal in C, there is $y \in I$ with $x \leq h(y)$. Then $x \leq \sup_L(h(I))$ for all $x \in C$. Therefore, $\sup_L(C) \leq \sup_L(h(I))$. Hence $\sup_L(C) = \sup_L(h(I))$.

As $h(I)$ is a chain in $[f \geq b]$, one has $\sup_L(C) = \sup_L(h(I)) \in [f \geq b]$, i.e.,

$$f(\sup_L(C)) \geq b > f(\sup_L(C)).$$

This contradiction proves (1). Then f is upward upper semicontinuous.

Similarly, f is downward upper semicontinuous if and only if for every $a \in \mathscr{C}$ and every chain C in $[f \geq a]$, one has $\inf_L(C) \in [f \geq a]$. The two parts finish the proof. □

Proposition 3.6 shows that the topological continuity is stronger than the order continuity. Combined with Theorem 6.8, it implies Corollary 6.14.

Proposition 3.6. *Assume that f is topologically upper semicontinuous. Then f is upper chain subcomplete. If further $\mathscr{C}$ is complete, then f is order upper semicontinuous.*

Proof. For every $a \in \mathscr{C}$, by the topological upper semicontinuity of f, the subset $[f \geq a] \subset L$ is closed. By Corollary 3.8, for every chain C in $[f \geq a]$, one has $\sup_L(C), \inf_L(C) \in [f \geq a]$. Therefore, f is upper chain subcomplete. When $\mathscr{C}$ is complete, by Proposition 3.5, f is order upper semicontinuous. □

We prove the results used in the proof of Proposition 3.6.

Lemma 3.7. *Let W, V be two nonempty subsets of L such that $C := W \cup V$ is a chain. Then $\sup_L(C) = \sup_L(W)$ or $\sup_L(C) = \sup_L(V)$.*

Proof. Since $W \subset C$ (resp. $V \subset C$), one has $\sup_L(W) \leq \sup_L(C)$ (resp. $\sup_L(V) \leq \sup_L(C)$). Hence $\sup_L(W) \vee \sup_L(V) \leq \sup_L(C)$. Since $\sup_L(W) \vee \sup_L(V)$ is an upper bound on C, one has $\sup_L(W) \vee \sup_L(V) = \sup_L(C)$. If for every $u \in W$, there is $v \in V$ with $u \leq v$, then $\sup_L(W) \vee \sup_L(V) = \sup_L(V)$. Otherwise, as C is a chain, there is $u_0 \in W$ such that $u_0 \geq v$ for all $v \in V$. Hence $u_0 \geq \sup_L(V)$ and $\sup_L(W) \vee \sup_L(V) = \sup_L(W)$. □

The interval topology of the lattice L is defined ([7, p.570]) to be the smallest topology such that $\{x \in L : x \geq a\}$ and $\{x \in L : x \leq a\}$ are closed for every $a \in L$. It is a topology canonically determined by the order structure. Roughly, Corollary 3.8 below means that a subset closed in this topology is also closed under certain order theoretic operations.

Corollary 3.8. *Let $A \subset L$ be a subset closed in the interval topology of L. Then A is chain-subcomplete in L.*

Proof. Let C be a nonempty chain in A. By the definition of the interval topology, one can write $A = \cap_{\alpha \in I} A_\alpha$, where for each index $\alpha \in I$, the subset $A_\alpha = \cup_{i=1}^{n(\alpha)} I_\alpha^i$, and for each $1 \leq i \leq n(\alpha)$, the subset I_α^i is a closed interval of L. In particular, for every $1 \leq i \leq n(\alpha)$, one has $\sup_L(I_\alpha^i \cap C) \in I_\alpha^i$. For every $\alpha \in I$, one has $C \subset A_\alpha$, so $C = \cup_{i=1}^{n(\alpha)}(I_\alpha^i \cap C)$. By Lemma 3.7, there is $1 \leq i_\alpha \leq n(\alpha)$ with $\sup_L(C) = \sup_L(I_\alpha^{i_\alpha} \cap C)$. For every $\alpha \in I$, one has $\sup_L(C) \in I_\alpha^{i_\alpha} \subset A_\alpha$. Therefore, one has $\sup_L C \in A$. Similarly, one has $\inf_L C \in A$. □

Remark 3.9. In Example 4.4, the subset $X \subset L$ is chain-subcomplete, but not closed in the interval topology. Whence, the converse of Corollary 3.8 fails. Topkis [20, Thm. 2.3.1] shows that for every integer $n \geq 0$, every chain-subcomplete *sublattice* of $\mathbb{R}^n$ is closed in the *Euclidean* topology. Still, the *subset* $\{(t, 1-t) | 0 < t < 1\} \subset \mathbb{R}^2$ is chain-subcomplete, but not closed in the Euclidean topology.

4 Quasisupermodular functions

The notion of quasisupermodular functions is a generalization of supermodular functions ([20, p.43]).

Definition 4.1. [13, p.162] A function $f : L \to C$ from a lattice to a chain is called *quasisupermodular* if for any $x, y \in L$, the condition $f(x) \geq f(x \wedge y)$ implies $f(x \vee y) \geq f(y)$, and the condition $f(x) > f(x \wedge y)$ implies $f(x \vee y) > f(y)$.

A function $g : L \to \mathbb{R}$ on a lattice is called supermodular if for any $x, y \in L$, one has[1] $g(x) + g(y) \leq g(x \wedge y) + g(x \vee y)$. Every supermodular function is quasisupermodular. Example 4.2 shows that unlike supermodular functions, the sum of two quasisupermodular function may not be quasisupermodular.

Example 4.2. Consider the sublattice $L = \{(1,1), (2,3), (3,2), (4,5)\}$ of $\mathbb{R}^2$. The function $f : L \to \mathbb{R}, \quad (x, y) \mapsto x$ is supermodular. The function $g : L \to \mathbb{R}, \quad (x, y) \mapsto -y$ is quasisupermodular. But the sum $h := f + g$ is not quasisupermodular, since $h((2,3) \wedge (3,2)) = h(1,1) < h(3,2)$ and $h(2,3) = h(4,5) = h((2,3) \vee (3,2))$.

Statement 4.3. [13, Theorem A3] Let L be a complete lattice. If $f : L \to \mathbb{R}$ is a quasisupermodular and order upper semicontinuous function, then f is upper semicontinuous in the interval topology of L.

However, Statement 4.3 is *false*[2] even when f is supermodular. Example 4.4 is a counterexample.

Example 4.4. Let X be an infinite anti-chain. Let L be the set obtained by adding two elements $m \neq M$ to X. We extend the partial order from X to L by requiring $m \leq x \leq M$ for every $x \in X$. Then L is a complete lattice. The interval topology of L is T_1, compact, but not Hausdorff.

Fix $x_0 \in X$. Define a function $f : L \to \mathbb{R}$ by setting $f(x) = 1$ for every $x \neq x_0$ and $f(x_0) = 0$. By Remark 3.3, since every chain in L is finite, f is supermodular and order upper semicontinuous. Moreover, $\arg\max f$ is compact but not closed in the interval topology of L. In particular, f is not upper semicontinuous in the interval topology. (Thus, the converse of Proposition 3.6 fails.)

[1] The condition uses the additive group structure of the codomain $\mathbb{R}$.

[2] Kukushkin [9, Example 3.2] provides a counterexample to [13, Theorem A2].

Take an element $x'(\neq x_0)$ of L. Then $f(x_0) < f(x')$ but f does not satisfy the defining property of transfer upper continuity. This appears to contradict the literal statement of [19, Theorem 2]. The reason is that the topology in [19, Theorem 2] is implicitly assumed to be Hausdorff.

5 Existence of maximum

In a normal form game, every player hopes to take best reply to other players' strategies. More precisely, every player wants to maximize his/her own payoff function within the set of available strategies. In this sense, the study of existence of a maximum of a given function is meaningful. The main objective of Section 5 is Theorem 5.3, which is used in Section 6.

Fact 5.1 is [19, Theorem 2], except that the authors implicitly assume X to be Hausdorff, as Example 4.4 explains. Since the interval topology of a complete lattice may not be Hausdorff, we state the following version. The original proof still works.

Fact 5.1 (Tian-Zhou)**.** *Let X be a compact (but not necessarily Hausdorff) topological space. Let $f : X \to \mathbb{R}$ be a transfer upper continuous function. Then* $\arg\max f$ *is a nonempty closed subset of X, hence compact.*

Let $f : L \to C$ be an upper chain subcomplete function from a complete lattice to a chain. Set $M := \arg\max f$. Proposition 5.2 shows that under an order-continuity hypothesis, order-theoretic property and topological property of the maximum points of a function are good. It generalizes [17, Propostion 3], which only asserts the completeness of M.

Proposition 5.2. *If M is a nonempty sublattice of L, then M is* subcomplete *in L. In particular, M is compact in the interval topology of L.*

Proof. For every nonempty chain $V \subset M$, since f is upper chain subcomplete and $M = \{s \in L : f(s) \geq \max f\}$, one has $\sup_L V, \inf_L V \in M$. By Theorem 2.3, the sublattice M is subcomplete in L. The second statement follows from [23, Theroem **??**]. □

Theorem 5.3 generalizes simultaneously [12, Theorems 1, 2] and [13, Thm. A4].[3] It relaxes the supermodularity condition to quasisupermodularity and strengthens the completeness result to subcompleteness. It gives a purely order-theoretic sufficient condition for the existence of maximum of functions on lattices. (A related equivalent condition is given in [10, Corollaries 3.1, 3.2].)

Theorem 5.3. *If further $f : L \to C$ is quasisupermodular, then* $\arg\max f$ *is a nonempty* subcomplete *sublattice of L.*

Proof. Define a correspondence $F : f(L) \to 2^L$ by $F(c) = \{x \in L : f(x) \geq c\}$. Since f is upper chain subcomplete, for every $c \in f(L)$, the value $F(c)$ is chain-subcomplete in L.

For any $c < c'$ in C, every $x \in F(c)$ and every $x' \in F(c')$ with $x \wedge x' \notin F(c)$, one has $f(x \wedge x') < c \leq f(x)$. Since f is quasisupermodular, one has $c' \leq f(x') < f(x \vee x')$. Thus, $x \vee x' \in F(c')$. Therefore, the correspondence F is increasing w.r.t the relation

[3] Although the proof of [13, Theorem A4] uses the wrong Statement 4.3, the statement itself is correct.

$\geq^{\mathrm{wV}}$ defined in [11, (2d)]. By [11, Theorem 2.2], there is an increasing selection $r : f(L) \to L$ of F.

For every $t \in f(L)$ and every $t' \in f(L)$ with $t' \geq t$, we have $r(t') \in F(t') \subset F(t)$. Hence, $\{r(t')\}_{t' \geq t}$ is a chain in $F(t)$. Since $F(t)$ is chain-subcomplete in L, the element $m := \sup_L r(f(L)) = \sup_L \{r(t')\}_{t' \geq t}$ exists and is in $F(t)$. Then $f(m) \geq t$ for every $t \in f(L)$. Therefore, f attains its minimum at m. In particular, $\arg\max f$ is nonempty.

We show that $\arg\max f$ is a sublattice of L. For any $u, v \in \arg\max f$, since $f(u \vee v) \leq \max f = f(u)$ (resp. $f(u \wedge v) \leq \max f = f(u)$) and f is quasisupermodular, one has $\max f = f(v) \leq f(u \wedge v)$ (resp. $\max f = f(v) \leq f(u \vee v)$). Therefore, $u \wedge v$ (resp. $u \vee v$) is in $\arg\max f$. By Proposition 5.2, the nonempty sublattice $\arg\max f$ is subcomplete. □

Remark 5.4. If the condition "upper chain subcomplete" of Theorem 5.3 is changed to "upper semicontinuity in the interval topology of L", then the sublattice $\arg\max f$ is compact in L. Then from completeness of L and [5, Footnote d, p.187], this sublattice is subcomplete. Still, Example 4.4 and Proposition 3.5 show that an upper chain subcomplete function may not be upper semicontinuous in the interval topology.

Zhou's theorem [24, Theorem 1] is stated for subcomplete-sublattice-valued correspondences. That is why we need Theorem 5.3 in the proof of Theorem 6.8.

6 Nash equilibria of quasisupermodular games

In Section 6, we recall the definition of quasisupermodular games and give several results about the existence and structure of the set of Nash equilibria.

Definition 6.1. [13, p.160] Let P, Q be two posets. Let C be a chain. A function $f : P \times Q \to C$ satisfies the single crossing property relative to (P, Q) if for any $p \lneq p'$ in P and any $q \lneq q'$ in Q, the condition $f(p, q) \leq f(p', q)$ implies $f(p, q') \leq f(p', q')$, and the condition $f(p, q) < f(p', q)$ implies $f(p, q') < f(p', q')$.

A function having increasing difference relative to $P \times Q$ (in the sense of [20, p. 42]) satisfies the single crossing property relative to (P, Q). However, unlike the increasing difference property, single crossing property is not symmetric in the two variables, as shown by [20, Example 2.6.9].

Lemma 6.2 compares the hypotheses of Theorem 6.8 with those in [21, Ch. 10, Theorem 2].

Lemma 6.2. *Let L, L' be two lattices. Let C be a chain. Let $f : L \times L' \to C$ be a function. Suppose that for every chain $\mathscr{C} \subset L'$, the restriction $f|_{L \times \mathscr{C}} : L \times \mathscr{C} \to C$ is quasisupermodular. Then f satisfies the single crossing property relative to (L, L'), and for every $t \in L'$, the function $f(\cdot, t) : L \to C$ is quasisupermodular.*

Proof. Consider $x \leq x'$ in L and $t \lneq t'$ in L'. Assume $f(x, t) \leq f(x', t)$ (resp. $f(x, t) < f(x', t)$). Then $\{t, t'\}$ is a chain in L'. Thus, $f|_{L \times \{t, t'\}}$ is quasisupermodular. Since $(x, t') \wedge (x', t) = (x, t)$ and $(x, t') \vee (x', t) = (x', t')$, one has $f(x, t') \leq f(x', t')$ (resp. $f(x, t') < f(x', t')$). Thus, f satisfies the single crossing property relative to (L, L').

Consider $t \in L'$. Then $\{t\}$ is a chain in L'. Thus, the function $f|_{L \times \{t\}} : L \times \{t\} \to C$ is quasisupermodular. Therefore, the function $f(\cdot, t) : L \to C$ is quasisupermodular. □

Example 6.3. Let $L = \{0, 1, a, b\}$ be a lattice, where $0 = \min L$, $1 = \max L$ and a, b are incomparable. Then $a \wedge b = 0$, $a \vee b = 1$. Let $T = \{0, 1\}$ be a *chain*. Define a function $f : L \times T \to \mathbb{R}$ by

$$f(0,0) = f(b,1) = 2, f(a,0) = f(b,0) = 1,$$
$$f(1,0) = f(1,1) = 0, f(a,1) = 9, f(0,1) = 10.$$

Then f satisfies the single crossing property relative to (L, T). For every $t \in T$, the function $f(\cdot, t) : L \to \mathbb{R}$ is quasisupermodular. However, f is not quasisupermodular. Indeed, one has $f((a,0) \wedge (b,1)) = f(0,0) = 2 = f(b,1)$ and $f((a,0) \vee (b,1)) = f(1,1) = 0 < 1 = f(a,0)$.

Definition 6.4. A quasisupermodular game

$$(N, \{S_i\}_{i \in N}, \{f_i\}_{i \in N}, \{C_i\}_{i \in N}) \tag{2}$$

is the following data:

1. a nonempty set of players N;
2. for every $i \in N$, a nonempty lattice S_i of the strategies of player i. We put $S = \prod_i S_i$ for the set of joint strategies and $S_{-i} := \prod_{j \neq i} S_j$;
3. for every $i \in N$, a nonempty chain C_i representing the possible gains of player i;
4. for each player $i \in N$, a payoff function $f_i : S \to C_i$ such that for every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to C_i$ is quasisupermodular (Definition 4.1);
5. for every $i \in N$, the function f_i satisfies the single crossing property relative to (S_i, S_{-i}).

Remark 6.5. Such a game bears various names in the literature. It is called a game with the single crossing property in [17, Sec. 3] and is said to have (ordinal) strategic complementarities in [13, p.175]. The definition on [20, p.179] requires further each S_i to be a sublattice of some Euclidean space.

From now to the end of the paper, fix a quasisupermodular game (2).

Definition 6.6 ((Pure) Nash equilibrium)**.** A joint strategy $x \in S$ is a Nash equilibrium of the game (2) if for every $i \in N$ and every $y_i \in S_i$, one has $f_i(y_i, x_{-i}) \leq f_i(x)$. The subset of S comprised of all Nash equilibria is denoted by E.

Remark 6.7. Although we cling to the analysis of pure-strategy Nash equilibria, mixed-strategy Nash equilibria are discussed in [6] for quasisupermodular games.

Classically, the best response correspondence is an essential tool to study Nash equilibria. For each player $i \in N$, the (individual) best response correspondence $R_i : S \twoheadrightarrow S_i$ is defined by

$$R_i(x) = \operatorname{argmax}_{y_i \in S_i(x_{-i})} f_i(y_i, x_{-i}).$$

The joint best response $R : S \twoheadrightarrow S$ is defined as $R(x) = \prod_{i \in N} R_i(x_{-i})$. Then E coincides with the set of fixed points of R.

Theorem 6.8 is purely order theoretic. For supermodular games, a purely order-theoretic result about the existence of the largest and the least Nash equilibria is in [12, p.1266].

Theorem 6.8. *Assume that for every $i \in N$, the lattice S_i is complete, and for every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to C_i$ is upper chain subcomplete. Then E (with the order structure inherited from S) is a nonempty complete lattice.*

Proof. By Theorem 5.3, for every $i \in N$, every $x \in S$, the value $R_i(x)$ is a nonempty subcomplete sublattice of S_i. Therefore $R(x)$ is nonempty subcomplete sublattice of S. Moreover, by [20, Theorem 2.8.6], the correspondence R_i is increasing in the sense of [20, p.33]. So R is also increasing. Because every S_i is complete, the product lattice S is complete. By Zhou's theorem [24, Theorem 1], E is a nonempty complete lattice. □

Remark 6.9. By Proposition 3.6, the continuity hypothesis of Theorem 6.8 is weaker than the topological upper semicontinuity.

Veinott [21, Ch. 10, Theorem 2] gives a similar result for parameterized games. When restricted to normal form games, Veinott's theorem requires that for every $i \in N$ and every chain $\mathscr{C} \subset S_{-i}$, the function $f_i|_{S_i \times \mathscr{C}} : S_i \times \mathscr{C} \to C$ is quasisupermodular. By Lemma 6.2, this hypothesis implies that the game is a quasisupermodular game in the sense of Definition 6.4. However, as Example 6.3 shows, a quasisupermodular game may not satisfy Veinott's hypothesis. Moreover, Veinott's theorem requires every C_i to be complete. In this sense, Theorem 6.8 is not covered by Veinott's theorem.

Example 6.10. Let $N = \{1, 2\}$. Let $C_i = \mathbb{R}$ for $i = 1, 2$. Let S_1 be the complete lattice L constructed in Example 4.4. Let S_2 be the finite lattice X in Example 6.3. Let x_0, f be as in Example 4.4. Define a function $f_1 : S_1 \times S_2 \to \mathbb{R}, \quad (s_1, s_2) \mapsto f(s_1)$. Then $f_1(\cdot, 0) : S_1 \to \mathbb{R}$ is not upper semicontinuous in the interval topology of S_1. Define a function $f_2 : S_1 \times S_2 \to \mathbb{R}$ by $f_2(s_1, 0) = 0$, $f_2(s_1, a) = f_2(s_1, b) = 1$ and $f_2(s_1, 1) = 1.5$ for every $s_1 \in S_1$. Then $f_2(s_1, \cdot) : S_2 \to \mathbb{R}$ is not supermodular. Still, the corresponding game satisfies the hypotheses of Theorem 6.8. In this case, the set $E = (L \setminus \{x_0\}) \times \{1\}$.

For every $i \in N$, assume $C_i = \mathbb{R}$. Let τ_i be a *compact* topology on S_i finer than the interval topology of S_i. By the Frink-Birkhoff theorem [1, Theorem 20, p.250], the lattice S_i is complete. Consequently, their product S is also a complete lattice.

Theorem 6.11 below is similar to [15, Theorem 1]. In [15, Theorem 1], each player's strategy set S_i is a chain, the set of players N is finite and only the existence of Nash equilibria is proved. On the one hand, we allow the S_i to be lattices, the set N to be infinite and the existence of the greatest Nash equilibrium is established. On the other hand, [15] weakens the single crossing condition (in Definition 6.4) to upward or downward transfer single-crossing condition ([15, Definition 4]).

Theorem 6.11. *Assume that for every $i \in N$, every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to \mathbb{R}$ is transfer weakly upper continuous relative to τ_i, and upward (resp. downward) upper semicontinuous. Then there is a largest (resp. least) Nash equilibrium.*

Proof. By symmetry, it suffices to prove the statement without parentheses. For every $i \in N$, every $x_{-i} \in S_{-i}$, by [19, Theorem 1], the compactness of τ_i and the transfer weak upper continuity of $f_i(\cdot, x_{-i})$ imply that $R_i(x)$ is nonempty. Therefore, by [20, Theorem 2.8.6], the correspondence $R_i : S \twoheadrightarrow S_i$ is increasing. So, the joint best response correspondence $R : S \twoheadrightarrow S$ is increasing and hence $R(x)$ is a sublattice of S for every $x \in S$.

We claim that for every $x \in S$, the maximum of $R(x)$ exists.

For every $i \in N$, every chain C in $R_i(x)$, the upward upper semicontinuity implies

$$f(\sup_{S_i}(C), x_{-i}) \geq \limsup_{t \in C, t \uparrow \sup_{S_i}(C)} f(t, x_{-i}) = \max_{S_i} f(\cdot, x_{-i}),$$

so $\sup_{S_i}(C)$ is in $R_i(x)$. Now that every chain in the nonempty poset $R_i(x)$ has an upper bound, by Zorn's lemma, $R_i(x)$ has a maximal element m_i. Then $(m_i)_i \in R(x)$ is a maximal element of $R(x)$. By Lemma 2.1, it is the maximum of $R(x)$. The claim is proved.

As R is increasing, the single-valued function $\max R : S \to S$ is increasing. As S is a complete lattice, $\max S$ exists. By Tarski's fixed point theorem [18, Theorem 1], the function $\max R : S \to S$ has a fixed point $x^* := \max\{y \in S : y \leq \max R(y)\}$. Then x^* is a Nash equilibrium.

We show that x^* is the largest Nash equilibrium. If $s \in S$ is another Nash equilibrium, then $s \in R(s) \leq \max R(s)$. By definition of x^*, we have $s \leq x^*$, which means that x^* is largest Nash equilibrium. □

Example 6.12. Let $N = \{1, 2\}$. Let $S_1 = [0, 1]$ and $S_2 = [0, 1]^2$. Define a function $f_1 : S_1 \times S_2 = [0, 1]^3 \to \mathbb{R}$ by

$$(s_1, a, b) \mapsto \begin{cases} 1 & s_1 > 0, \\ 0 & s_1 = 0. \end{cases}$$

Then $f_1(\cdot, 0, 0) : S_1 \to \mathbb{R}$ is not order upper semicontinuous. Define a function $f_2 : S_1 \times S_2 = [0, 1]^3 \to \mathbb{R}, \quad (s_1, a, b) \mapsto s_1 + a + b$. Then the resulting game satisfies the conditions of Theorem 6.11. The set E is $(0, 1] \times \{(1, 1)\}$. The largest Nash equilibrium is $(1, 1, 1)$, but there is no least Nash equilibrium. As S_2 is not a chain, one cannot apply [15, Theorem 1] in this case.

With a topological hypothesis (on the payoff functions) which is strictly weaker than the continuity, we can get the same order structure of the set of Nash equilibria as in Theorem 6.8.

Theorem 6.13. *Assume that for every $i \in N$ and every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to \mathbb{R}$ is transfer upper continuous relative to τ_i. Then E is a nonempty complete lattice.*

Proof. For every $i \in N$ and every $x \in S$, by Fact 5.1 and the compactness of τ_i, the value $R_i(x)$ is nonempty and compact in (S_i, τ_i). As τ_i is finer than the interval topology of S_i, $R_i(x)$ is also compact in the interval topology. By [20, Theorem 2.8.6], the correspondence $R_i : S \twoheadrightarrow S_i$ is increasing. Therefore, the correspondence $R : S \twoheadrightarrow S$ is increasing and $R_i(x)$ is a sublattice of S_i. By [23, Theorem **??**], the sublattice $R_i(x)$ is subcomplete in S_i. Therefore, $R(x)$ is a nonempty subcomplete sublattice of S. By Zhou's theorem [24, Theorem 1], E is a nonempty complete lattice. □

When N is finite, Corollary 6.14 specializes to Calciano's result in [2, Sec 4.1] about the structure of the set of Nash equilibria.

Corollary 6.14 (Calciano)**.** *Suppose that for every $i \in N$ and every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to \mathbb{R}$ is upper semicontinuous relative to τ_i. Then E is a nonempty complete lattice.*

Proof. It follows from Theorem 6.13. □

Remark 6.15. Under an extra continuity hypothesis (namely, for every $i \in N$ and every $x_i \in S_i$, the function $f_i(x_i, \cdot) : S_{-i} \to \mathbb{R}$ is continuous), the existence of both least and largest Nash equilibria is showed in [13, Theroem 12], extending a previous result [17, Proposition 4].

Example 6.16. Let $N = \{1, 2\}$. Let $S_1 = S_2 = [0, 1]$. Define a function $f_1 : S_1 \times S_2 \to \mathbb{R}$ by

$$(s_1, s_2) \mapsto \begin{cases} 0 & s_1 \leq 1/3, \\ 1 & 1/3 < s_1 < 2/3, \\ 2 & 2/3 \leq s_1. \end{cases}$$

Then $f_1(\cdot, 0) : S_1 \to \mathbb{R}$ is not upper semicontinuous. Define $f_2 : S_1 \times S_2 \to \mathbb{R}, \quad (s_1, s_2) \mapsto (s_1 + 1)(s_2 - s_2^2)$. The resulting game satisfies the conditions of Theorem 6.13 but not those of Corollary 6.14. The set $E = [2/3, 1] \times \{1/2\}$. Thus, Theorem 6.13 is strictly stronger than Corollary 6.14.

Corollary 6.17. *If for every $i \in N$, the set S_i is finite, then E is a nonempty complete lattice.*

Proof. For every $i \in N$, take τ_i to be the discrete topology on S_i. Then τ_i is compact, and for every $x_{-i} \in S_{-i}$, the function $f_i(\cdot, x_{-i}) : S_i \to \mathbb{R}$ is upper semicontinuous relative to τ_i. We conclude by Corollary 6.14. □

Statements and Declarations

Conflict of interests

The author declares that she has no conflict of interests.

Ethical approval

This article does not contain any studies with human participants or animals performed by the author.

Informed consent

For this type of study informed consent was not required.

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